Thus, Einstein and Newton, with their very different viewpoints on the nature of space and time, give very different names to the agent that causes the crossing. Einstein calls it spacetime curvature; Newton calls it tidal gravity. But there is just one agent acting. Therefore, *spacetime curvature and tidal gravity must be precisely the same thing, expressed in different languages.*

Our human minds have great difficulty visualizing curved surfaces with more than two dimensions; therefore, it is nearly impossible to visualize the curvature of four-dimensional spacetime. Some insight can be gained, however, by looking at various two-dimensional pieces of spacetime. Figure 2.6 uses two such pieces to explain how spacetime curvature creates the tidal stretch and squeeze that produce the ocean tides.

Figure 2.6a depicts one piece of spacetime in the vicinity of Earth, a piece that includes time, plus space along the direction toward the Moon. The Moon curves this piece of spacetime, and the curvature stretches apart two geodesics, in the manner shown. Correspondingly,we humans see two freely moving particles, which travel along the geodesics, get stretched apart as they travel, and we interpret that stretching as a tidal gravitational force. This stretching tidal force (spacetime curvature) affects not only freely moving particles, but also the Earth's oceans; it stretches the oceans in the manner shown in Box 2.5, producing oceanic bulges on the sides of the Earth nearest and farthest from the Moon. The two bulges are trying to travel along geodesics of the curved spacetime (Figure 2.6a), and therefore are trying to fly apart; but the Earth's gravity (the spacetime curvature produced by the Earth; not shown in the diagram) is counteracting that flight, so the ocean merely bulges.

Figure 2.6b is a different piece of spacetime near Earth, a piece that includes time, plus space along a direction transverse to the Moon's direction. The Moon curves this piece of spacetime, and the curvature squeezes geodesics together in the manner shown. Correspondingly, we humans see freely moving particles that travel along geodesics transverse to the Moon's direction get squeezed together by the curvature (by the Moon's tidal gravity), and similarly we see the Earth's oceans get squeezed along directions transverse to the direction of the Moon. This tidal squeeze produces the transverse oceanic compressions shown in Box 2.5.